

VALERI MASCARILLA

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SUMMARY

Mechanical and aerospace engineer (M.S., GPA 4.0) with nearly 2 years in aerospace implementing drawing changes and design change requests, analyzing drop-test data, and validating designs with FEA (Nastran) and CFD (OpenFOAM) in SolidWorks and CATIA V5. U.S. work authorization: F-1 OPT from Jan. 2027, STEM OPT eligible; no sponsorship needed to start.

EDUCATION

Illinois Institute of Technology Jan. 2026 – Dec. 2026
Master of Science, Mechanical and Aerospace Engineering; GPA: 4.0/4.0
Chicago, IL

Relevant coursework: Turbulent Flows; Dynamics of Compressible Fluids; Data-Driven Modeling

Universitat Politècnica de Catalunya Sep. 2023 – Jun. 2025
Master of Science, Aeronautical Engineering (dual degree with Illinois Tech, expected Dec. 2026)
Barcelona, Spain

Relevant coursework: Production and Design Aerospace; Aerospace Materials; Computational Engineering

Universitat Politècnica de Catalunya Sep. 2019 – Jun. 2023
Bachelor of Science, Aerospace Vehicle Engineering
Barcelona, Spain

EXPERIENCE

Aerospace Engineer Feb. 2024 – Dec. 2025
CIMSA Ingeniería de Sistemas, S.A.
Barcelona, Spain

- Implemented engineering drawing changes and design change requests for parachute assemblies, applying GD&T
- Analyzed instrumentation and sensor data from more than 20 drop tests to isolate root causes and define design changes
- Resolved nonconformances with suppliers and production through root cause analysis and supported FMEA
- Authored test plans, test reports and quality documentation to AS9100, AQAP-2110 and ECSS for client design reviews
- Managed manufacturing of 3 simultaneous programs exceeding \$1M each, coordinating production and inspection staff

PROJECT EXPERIENCE

Lightweight Telescope Mirror | SolidWorks, ANSA, MSC Nastran, META Spring 2026

- Designed a space structure from concept to validated model against stiffness, mass, and interface requirements
- Reduced structural mass by 14% through iterative design optimization
- Ran modal and static FEA in MSC Nastran to verify structural integrity and document margins of safety

Modeling Turbulence with AI | OpenFOAM, Python, PyTorch, MATLAB Jan. 2026 – Present

- Generated CFD training data in OpenFOAM and built machine learning models that predict flow from limited data
- Reduced computational cost of flow simulations by orders of magnitude through surrogate modeling
- Developed a physics-informed synthetic-data method that improved surrogate model accuracy on limited datasets

Aircraft Wing Structural Analysis | MATLAB, Finite Element Analysis (FEA) Spring 2025

- Developed a finite element analysis (FEA) solver from scratch to analyze a wing structure under flight loads
- Compared modeling approaches to assess accuracy against computational cost for early design iterations

SKILLS

CAD: SolidWorks | CATIA V5 | Engineering drawings | GD&T | Drawing changes and design change requests

Simulation: FEA: ANSA | MSC Nastran | META | CFD: OpenFOAM | Ansys Fluent

Programming: Python | MATLAB | C++ | Git

Engineering: Root cause analysis | Test planning | FMEA | Nonconformance | AS9100 | Project management | ERP

Languages: Spanish (Native) | Catalan (Native) | English (Professional) | German (Intermediate)

VOLUNTEER EXPERIENCE

Volunteer Firefighter May 2023 – Present
Catalonia's government fire department
Lleida, Spain

- Trained in emergency response, wildfire suppression, and rescue operations; 50+ emergencies a year
- Operate in teams of 3 to 5 people in high-risk, safety-critical environments requiring rapid decision-making and teamwork